

Beal's conjecture disproof

Problem

If $A^x + B^y = C^z$, where A, B, C, x, y and z are positive integers and x, y and z are all greater than 2, then A, B and C must have a common prime factor

Abstract

An observation. I got confused because of the “if” word on problem. If $x = y$, means if they are equal, not whenever they are equal. So I thought that by proving that $x \neq y$, would disproof. But it doesn't. So I need to prove that condition doesn't hold for any number.

$$A^x + B^y = C^z$$

Divide both sides by C^z

$$\frac{A^x}{C^z} + \frac{B^y}{C^z} = 1$$

Divide both sides by A^x

$$\frac{A^x}{C^z * A^x} + \frac{B^y}{C^z * A^x} = \frac{1}{A^x}$$

Simplify left side

$$\frac{A^x + B^y}{C^z * A^x} = \frac{1}{A^x}$$

Put A^x into evidence on left side

$$\frac{A^x(1 + \frac{B^y}{A^x})}{C^z * A^x} = \frac{1}{A^x}$$

Simplify equation on left side

$$\frac{\cancel{A^x}(1 + \frac{B^y}{A^x})}{C^z * \cancel{A^x}} = \frac{1}{A^x}$$

$$\frac{(1 + \frac{B^y}{A^x})}{C^z} = \frac{1}{A^x}$$

Multiply C^z by both sides

$$(1 + \frac{B^y}{A^x}) = \frac{C^z}{A^x}$$

We will now, try to prove that this isnt equal.

$$\frac{C^z}{A^x} > \frac{B^y}{A^x + 1}$$

Multiply both sides by $\frac{1}{B^y}$

$$\frac{C^z}{A^x * B^y} > \frac{1}{A^x + 1}$$

Multiply both sides by $\frac{1}{C^z}$

$$\frac{1}{A^x * B^y} > \frac{1}{C^z(A^x + 1)}$$

Develop denominator on right side

$$\frac{1}{A^x * B^y} > \frac{1}{C^z * A^x + C^z}$$

Assuming

$A^x + B^y = C^z$, we can conclude that C^z is bigger than A^x or B^y

Given

$$C^z * A^x + C^z > A^x * B^y,$$

denominator of right side bigger. Means its smaller number than left side

So inequation holds. So

$$(1 + \frac{B^y}{A^x}) \neq \frac{C^z}{A^x}$$

not equal .

Beal's Conjecture doesn't holds.

Remarks

Intervals

If we want to know what interval does a number belongs within. e.g

$$\text{factor} * \text{factor} * \text{factor} < \text{number} < \text{factor2} * \text{factor2} * \text{factor2}$$

Suppose I want to know which interval does 152 belongs

We factorize 152, in primes. We get $2^3 * 19$. So

$$2*2*2 < 152 < 2*2*2*19$$

We don't get $2*2*2*19$, because 19, is a prime, and can't be factorized...

Parity used

Another thing revelant, is that we can use parity property, to exclude possibilites.

We can't have

$$\text{even} + \text{odd} = \text{even} \text{ (not possible)}$$

Try to get all solutions

Supposing that p is a prime, and equal to A (on problem). So, we get

$$p^x + B^y = C^z$$

Say A and B, also are that prime. Then we get

$$p^x + p^y = C^z$$

We can also use composites. Doesnt matter, if the composite is on form $(\text{prime1} * \text{prime2})^z$, or $(\text{prime1} * \text{prime2} * \text{prime3})^z$. Suppose the base, is W. Then we get W^z .

E.G:

$$p^x + B^y = (W * p)^z \text{ (in this case } C^z, \text{ share a factor with } A^x, p)$$

Also:

$$p^x + B^y = (W)^z \text{ (in this case, it doesn't share factor with } A^x)$$

In the below formula, p is a prime. W is a composite. They share factor p .
Supposing $p^x < p^y$

$$p^x + p^y = (Wp)^z$$

Put p^x in evidence on left side

$$p^x(1 + p^{y-x}) = (Wp)^z$$

Develop right side

$$p^x(1 + p^{y-x}) = W^z * p^z$$

This results in

$$\frac{p^x}{p^z} \times (1 + p^{y-x}) = W^z$$

This is a good starting equation, to try to find all solutions for problem. (for example, for z bigger than x , we can eliminate some factors, especially, if , in this case $x > z$)

To put in evidence (like in the above formula), I advise you to always put in evidence the smaller number, to avoid having fractions. So we put A^x in evidence because x , is smaller than y .

Numbers on pure power form

It is also useful if we know ,that a number is on the form of pure power, as in
number = $x * x * x * x ..$ For example for the formula

$$\frac{p^x}{p^z} \times (p^{x-y} + 1) = W^z$$

Is useful to know if $p^{x-y} + 1$, on form of pure power, or not..

Is $1-A^{\{x-z\}}$ on form $A^*A^*A^*...$?

For $x > z$

we get

$$1 - A^{\{\text{exponent positive}\}}..$$

If we have A^x , previous number on a^*a^*a , form is $(A^{\{x-1\}})$

$A^{\{x-1\}}$ can't be one. So no not in the form.

For $x=z$

$1-1$, gives zero...not on form A^*A^*A

For $z > x$

$1 - 1 / A^{\{\text{exponent}\}}$, smaller than one. Not on form $A^*A^*A...$

KeyWords: divisibility, primes, number theory

André Albergaria Coelho

Email : andrealbergaria@gmail.com

Date : 2025 / 08 / 18